Project Progress Report — Week 11: Adversarial Stress Testing & Edge Cases

Work done in current Week:

- Conducted adversarial stress testing against Sybil attacks and rapid device ID spoofing.
- Verified that Rule Engine catches borderline cases where GNN confidence dips below 0.50.

Work to be done in next Week :

- Integrate GNNExplainer for subgraph visualization and nodal feature importance.

Supervisor Comments:

Adversarial evaluation confirms robustness against evasion attempts by fraud syndicates.

Supervisor Signature/Date: .....

Project Progress Report — Week 12: Explainable AI (XAI) & Interpretability

Work done in current Week:

- Implemented GNNExplainer integration to extract 2-hop topological attribution subgraphs.
- Mapped model decisions to human-readable compliance tags for banking and RBI audit trails.

Work to be done in next Week :

- Execute multi-threaded concurrency and high-throughput stress testing.

Supervisor Comments:

XAI features provide strong regulatory compliance for financial decision audits.

Supervisor Signature/Date: .....

Project Progress Report — Week 13: High Concurrency & Load Testing

Work done in current Week:

- Benchmarked system under simulated high load of 1,000 transactions/second on multi-core CPU.
- Maintained median P50 latency of 5.8 ms with zero dropped transactions or memory leaks.

Work to be done in next Week :

- Compile comprehensive project thesis, IEEE format research paper, and user guide.

Supervisor Comments:

Throughput and memory stability meet commercial payment processing criteria.

Supervisor Signature/Date: .....

Project Progress Report — Week 14: Research Paper Drafting & Documentation

Work done in current Week:

- Drafted research paper titled 'Real-Time UPI Fraud Detection Using Inductive GraphSAGE'.
- Compiled complete technical thesis chapters covering methodology, loss formulas, and benchmarks.

Work to be done in next Week :

- Conduct mock viva presentations and practice interactive UI demo workflows.

Supervisor Comments:

Documentation and research paper draft are well-structured and technically rigorous.

Supervisor Signature/Date: .....

Project Progress Report — Week 15: Mock Viva Defense & Final Submission

Work done in current Week:

- Rehearsed live examiner demo covering Auto-Stream, clean transactions, and mule ring attack.
- Finalized project deliverables, code repository archiving, and supervisor sign-offs.

Work to be done in next Week :

- Appear for final external project viva evaluation and committee defense.

Supervisor Comments:

Excellent execution across all engineering milestones. Fully approved for final examination.

Supervisor Signature/Date: .....